

LINUX LAB-1

Que-3

Que 3: Create a College directory hierarchy with subdirectories (CSE, ECE, ME, Civil), nested semester/subject folders, and display it using tree, pwd, cd, mkdir, rmdir, ls.

Ans 3: Linux provides commands for creating, navigating and displaying directories. The mkdir command creates directories, while the -p option allows creation of missing parent directories. The cd command is used to navigate between directories and pwd displays the current working directory. The tree command displays the complete directory hierarchy in a visual tree format. The ls command lists directory contents, while rmdir removes an empty directory. These commands were used to create and verify the College directory hierarchy containing CSE, ECE, ME and Civil departments with nested semester and subject folders.

Result

The College directory hierarchy was successfully created with CSE, ECE, ME and Civil departments, along with nested

semester and subject directories. The hierarchy was verified using tree, pwd and ls, and an empty directory was successfully removed using rmdir.

```
root@luffy: /home/LabFiles  x  +  v
root@luffy:/home/LabFiles# mkdir -p College/{CSE,ECE,ME,Civil}/{Sem1,Sem2}/{Subject1,Subject2}
root@luffy:/home/LabFiles# pwd
/home/LabFiles
root@luffy:/home/LabFiles# cd College/CSE/Sem1
root@luffy:/home/LabFiles/College/CSE/Sem1# pwd
/home/LabFiles/College/CSE/Sem1
root@luffy:/home/LabFiles/College/CSE/Sem1# cd -
/home/LabFiles
root@luffy:/home/LabFiles# tree College
College
├── CSE
│   ├── Sem1
│   │   ├── Subject1
│   │   └── Subject2
│   └── Sem2
│       ├── Subject1
│       └── Subject2
├── Civil
│   ├── Sem1
│   │   ├── Subject1
│   │   └── Subject2
│   └── Sem2
│       ├── Subject1
│       └── Subject2
├── ECE
│   ├── Sem1
│   │   ├── Subject1
│   │   └── Subject2
│   └── Sem2
│       ├── Subject1
│       └── Subject2
└── ME
    ├── Sem1
    │   ├── Subject1
    │   └── Subject2
    └── Sem2
        ├── Subject1
        └── Subject2

29 directories, 0 files
root@luffy:/home/LabFiles# rmdir College/Civil/Sem2/Subject2
root@luffy:/home/LabFiles# ls College
CSE  Civil  ECE  ME
root@luffy:/home/LabFiles# ls -R College | head -20
College:
CSE
Civil
ECE
ME

College/CSE:
Sem1
Sem2

College/CSE/Sem1:
Subject1
Subject2

College/CSE/Sem1/Subject1:

College/CSE/Sem1/Subject2:

College/CSE/Sem2:
Subject1
root@luffy:/home/LabFiles# |
```